AGENT FLEET BLUEPRINT — the complete finalist plan (non-LLM trading fleet)

Session 28 · 2026-10-27 · final blueprint for owner review Scope: every decision locked through Sessions 22–28 — 75 symbol agents, per-market styles, architecture, end-to-end workflow, rules & regulations, training / self-improvement, data requirements, latency + operations-per-second budget, build waves with time estimates. **Status:** draft v1 → **council-validated** (§14) → final. Every claim classified: measured / web-verified / design / unverified. **Diagrams:** `diagrams/d1-architecture.png` … `d6-e2e.png` (6 figures, embedded below).

---

§1 · The fleet at a glance

Unit	Count	What it is
Agents (brains)	**75**	one research agent + one regime router per symbol (Mode R, offline)
Cells (hands) — ceiling	~460–520	one executor per (symbol × style), built only if E4 promotes it
Active at once	**≤ 75**	router enables **0 or 1** style per symbol; default = all off
Day-1 live cells	2–4	wave 1 = index options first, then liquid stock options

Market	Symbols	Count	Styles/symbol	Cell ceiling
Stocks (Upstox NSE/BSE, options BUYING only, same-day)	NIFTY + SENSEX + optionable weight-stocks (scan S3)	**≤ 59**	8–9 (S1: MM conditional)	~440–500
FX Team 1 (cTrader ECN)	7 pairs + Brent/WTI/XAU/XAG	**11**	3 pairs (session·carry·news) / 2 commodities	**29**
FX Team 2 (cTrader, **separate account**, pure scalping)	XAU/USD + WTI	**2**	1 (scalping only, push-mode)	**2**
Crypto (Hyperliquid perps)	BTC, ETH, HYPE	**3**	4 (funding·liquidation·weekend·PAIR)	**10–12**
Total		**75**		**≈ 480–545** (recomputed Session 33: 481–543)

Unit clarification (owner Q, Session 28): 73 (old) was symbols × 1 style. The multiplication was always in the plan (O5: "2 styles = 146"). Brains = 75; hands = many; **only one hand works per symbol at a time (G5)**; most cells are never built (only E4-promoted styles exist).

---

§2 · Symbols and styles — the locked rosters

2.1 Stocks (≤59 symbols, options BUYING only)

- **Scan rule (S3):** universe = NIFTY 50 weight ∪ SENSEX 30 weight ∪ option universe → one list **≤ 59**. This scan IS the D4 screen; non-optionable names drop (may reduce the 57).
- **Instruments:** options only, **buy side only** (sellers post full premium + margin — the owner's verified reason). **Fully same-day: buy and sell same day / minutes–hours, never hold (S2)** → theta negligible; **spread cost is the battle**.
- **Index workhorse:** NIFTY (NFO) + SENSEX (BFO) weeklies carry the intraday load; stock options (monthly, thinner) serve event/swing cells.
- **Styles (8–9):** intraday-MF · stat-arb PAIR · earnings/event · auction · **equity MM (S1: adapts by style condition — conditional/gated, not deleted)** ·
  - option styles delta-direction · vol-alpha · gamma-scalper · hedge-executor. **Council fix Q10: hedge-executor is a portfolio overlay (cost), not an alpha style** — long protective puts have negative standalone expectancy by construction, so it is **governed by E5 (hedge effectiveness), excluded from E4 alpha promotion.**
- **Same-day research power (council fix Q12):** day-1 statistical claims restricted to NIFTY/SENSEX weeklies (only adequately sampled instrument); stock-option cells pool symbols hierarchically with a shared cost/entry model.
- **Hierarchy:** 57 weight-stock agents → **NIFTY agent (NFO)** / **SENSEX agent (BFO)**; compressed "important-only" comms both ways (S4). Index agents never edit cell artifacts (G1).

2.2 FX Team 1 — 11 symbols, one agent per symbol, offline + online

Group	Symbols	Styles	Cells
7 pairs	EUR/USD, GBP/USD, USD/JPY, AUD/USD, USD/CAD, USD/CHF, USD/SGD	session-pattern · carry · news-fade	21
4 commodities	Brent, WTI, XAU/USD, XAG/USD	session-pattern · news-fade (no carry)	8

Per symbol: 1 research agent (Mode R: hypothesis · GP · ML · alt-data · microstructure) → E4 gate → specialist cells (Mode T). Router: session→session-pattern, rate-diff→carry, calendar→news rules, ambiguous→**all off**.

2.3 FX Team 2 — gold + WTI only, separate account, pure scalping

Item	Design
Symbols	**XAU/USD + WTI only** (2 cells), **separate account** — scalping risk isolated
Style	**1 scalping style only** — one agent, one symbol, one style, push-mode
Two-mode	offline learns *this symbol's* microstructure; online executes every tick
Machinery	Session 25 plan: NY4 VPS, **maker-entry mandatory**, cost floor per tick
Exits	hard tick stop · spread-widening gate · EIA/CPI print pre-flatten (oil) · kill-switch

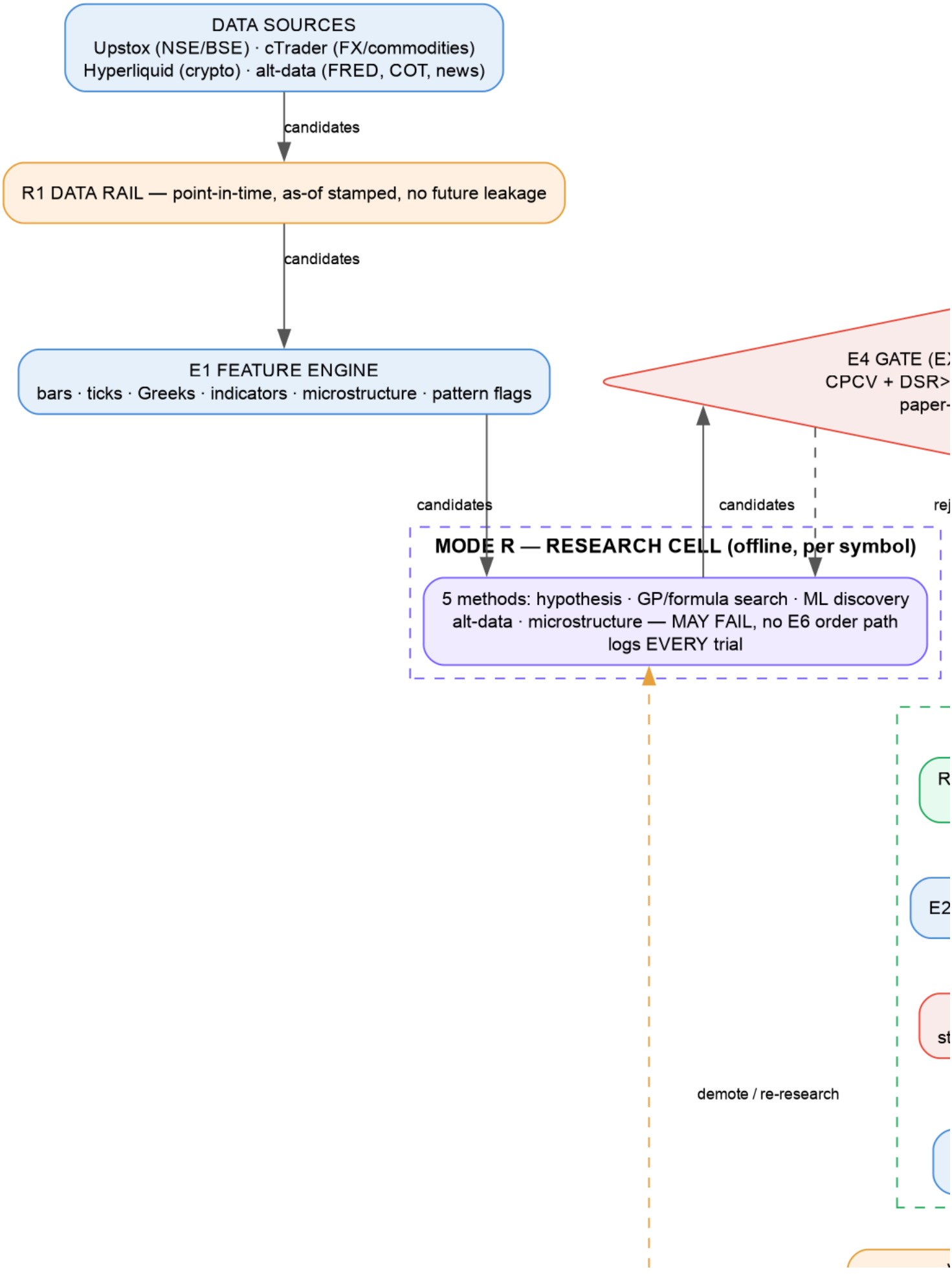
Push honesty: Mode R pushes the *search* hard; **E7 proves net-of-cost per attempt → E5 allows size only while net edge is positive. The machine pushes; the governor decides how far.**

2.4 Crypto — BTC, ETH, HYPE (Hyperliquid perps)

- **Tradeable styles: 4 of 6** — funding-rate · liquidation-cascade · weekend-quirks · stat-arb PAIR. ❌ cross-venue MM + perp MM = **deferred** (Hyperliquid rebate tier unreachable).
- Cells: 3 symbols × 3 styles = 9 + 1–3 PAIR = **10–12**.
- Exits: cascade-flatten · weekend-gap guard · funding-window (00/08/16 UTC) = the "print" · kill-switch. 24/7 market, no theta.

---

§3 · Architecture — rails, engines, orchestration


DATA SOURCES
Upstox (NSE/BSE) · cTrader (FX/commodities)
Hyperliquid (crypto) · alt-data (FRED, COT, news)
candidates
R1 DATA RAIL — point-in-time, as-of stamped, no future leakage
candidates
E1 FEATURE ENGINE
bars · ticks · Greeks · indicators · microstructure · pattern flags
E4 GATE (E
CPCV + DSR>
paper-
candidates
candidates
rej
MODE R — RESEARCH CELL (offline, per symbol)
5 methods: hypothesis · GP/formula search · ML discovery
alt-data · microstructure — MAY FAIL, no E6 order path
logs EVERY trial
R
E2
st
demote / re-research

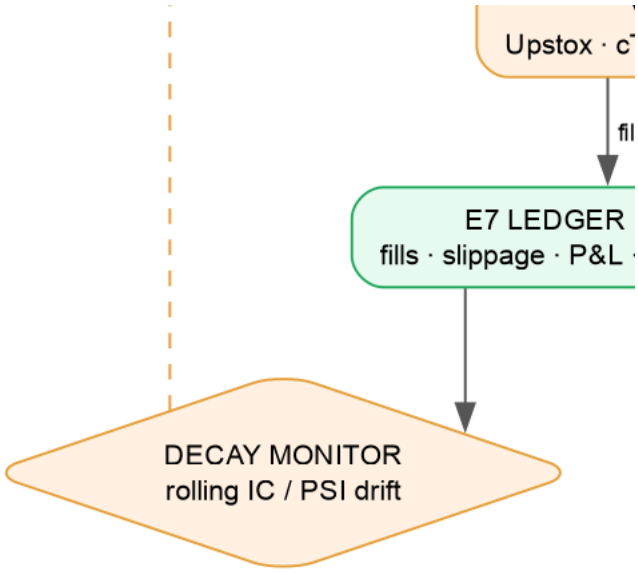
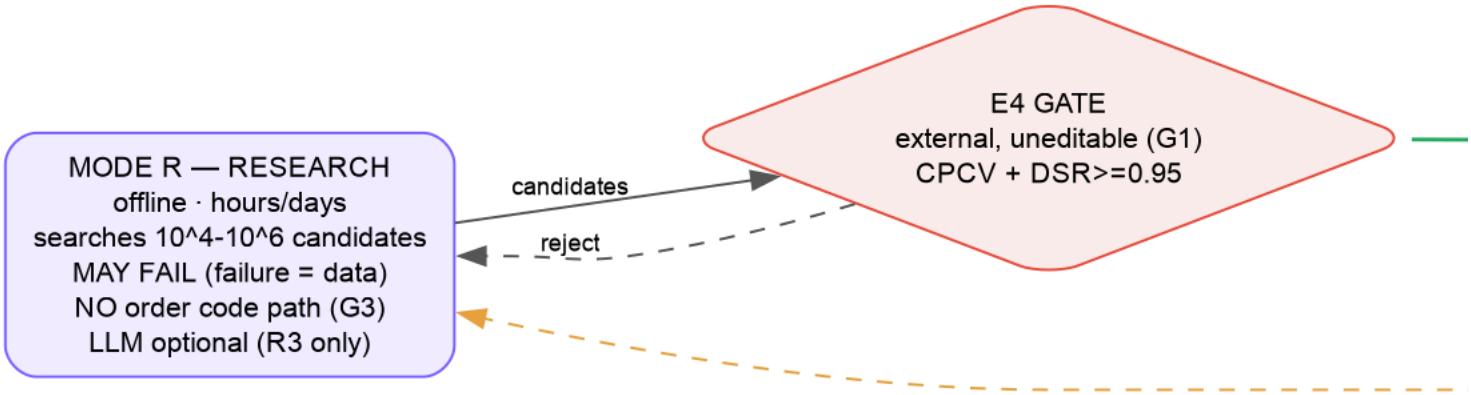


Figure D1 — the machine. Data → R1 rail → E1 features → Mode R research → E4 gate → R3 registry → Mode T (router → E2 signal → E5 governor → E6 execution) → venues → E7 ledger → decay monitor → back to research.

Layer	Name	Job	Class
Rails	R1 DATA	point-in-time, as-of stamped bars/ticks/Greeks/alt-data — **no future leakage**	design
	R2 SIGNAL	frozen formulas/ONNX only in the hot path	design
	R3 LEARNING	offline only — LLM allowed **here only**, never in R1/R2 (R3 non-negotiable)	design
Engines	E1 features	bars · ticks · Greeks · indicators · microstructure · pattern flags	design
	E2 signal	evaluates frozen artifacts each tick	design
	E4 gate	**external, uneditable** — CPCV + **DSR ≥ 0.95** (probability metric, council fix C1) + crisis replay + net-of-cost → paper → shadow 3–6 mo; **owner approves each promotion, ≤5 decisions/week**	design
	E5 governor	deterministic: stops · size caps · news ladder · kill-switch · ban checks	design
	E6 execution	TWAP/VWAP/limits · venue adapters (Upstox / cTrader / Hyperliquid)	design
	E7 ledger	fills · slippage · P&L · **costs in the label before promotion**	design
Orchestration	L0–L4	cell → symbol agent → team → market → global governor (kill-switch cascades down)	design

3.1 One cell, two modes (Session 26)

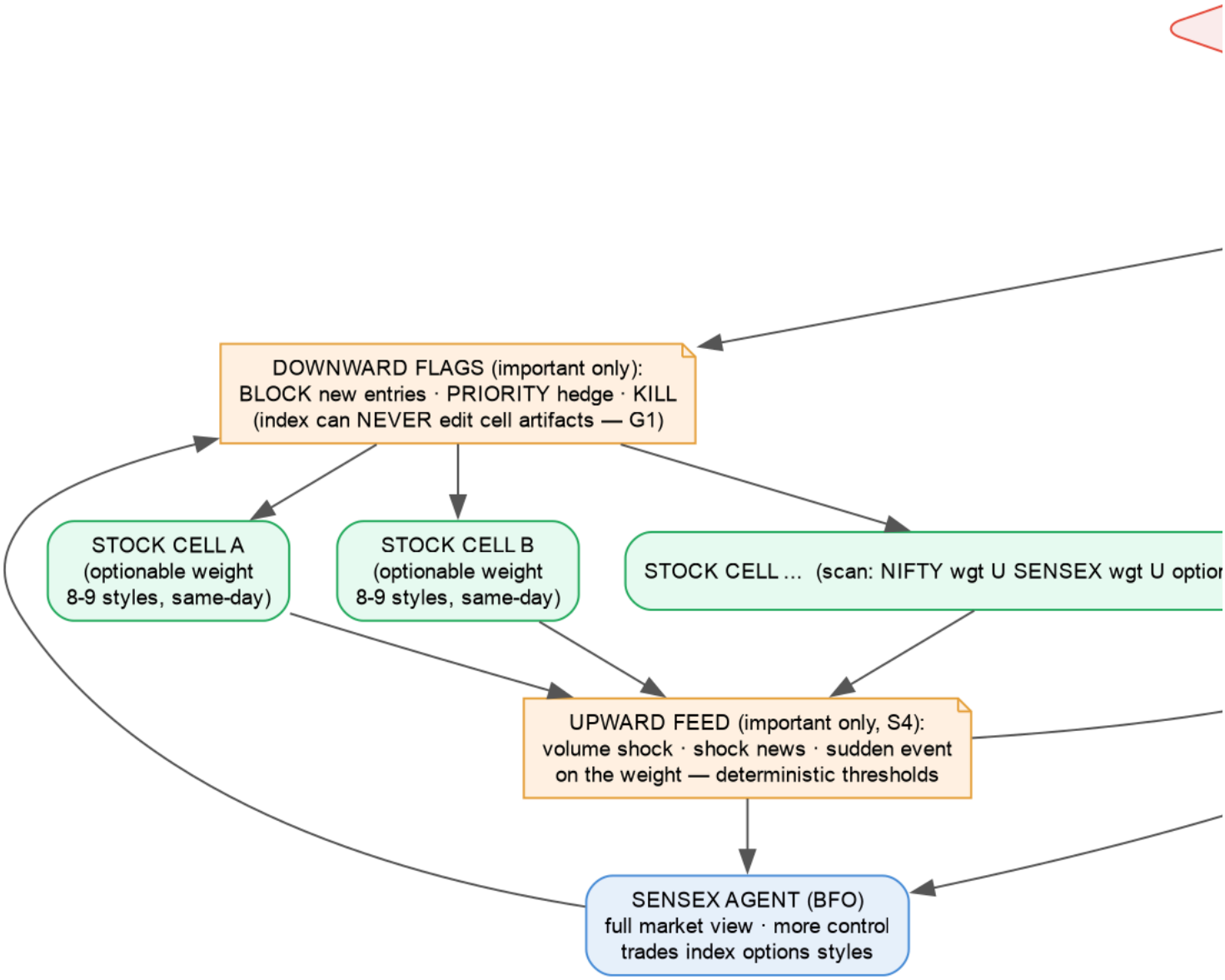


	Mode R (research, offline)	Mode T (trading, online)
Runs	hours–days, per symbol	us–ms, per tick
May	search 10^4–10^6 candidates, **may fail** (failure = data)	execute promoted artifacts only
Must not	hold order code path (**G3**)	contain search code (**G2**) or edit itself (**G1**)
LLM	optional (R3 only)	**never**
Isolation	separate process (**G6**)	separate process (**G6**)

Guards G1–G6: G1 cells can't edit their own artifacts · G2 no search code in Mode T · G3 research has no order path · G4 every trial logged (DSR denominator) · G5 one style active per symbol · G6 Mode R/T separate processes.

---

§4 · The stocks system (≤59 agents as ONE system)



Owner's S1–S4 answers (locked):

#	Decision	Effect
S1	equity MM **adapts by style condition**	MM stays, gated — up to 9 styles/symbol
S2	fully same-day buying+selling, **never hold**	all stock styles intraday; theta ≈ 0; spread cost = the battle
S3	scan NIFTY wgt ∪ SENSEX wgt ∪ option universe → **one list ≤ 59**	this scan IS the D4 screen; may drop some of 57
S4	upward comms = **volume shock + shock news + sudden event** on the weight (deterministic thresholds)	weights move the index — rules-only, R2-safe

Comms protocol (important-only, both directions):

- **Stock → index:** sends only on threshold crossing (regime flip / style enable / kill participation) — triggered by new volume flow, shock news, sudden events on that weight.
- **Index → stock:** only `BLOCK` new entries · `PRIORITY` hedge · `KILL` participation.
- **Index agents NEVER edit** stock cells' frozen artifacts (G1 intact).
- NIFTY agent (NFO) has the aggregated bottom-up view + more control; same for SENSEX (BFO).

---

§5 · How each market's agent works

5.1 FX Team 1 (11 symbols — steady rotation)

Phase	What happens
Offline (Mode R)	per symbol: hypothesis → GP/ML search → alt-data (COT, rate diffs) → microstructure; 5 methods; every trial logged
Gate	E4: CPCV + DSR ≥ 0.95 + net-of-cost → frozen artifact in R3
Online (Mode T)	router: session state → session-pattern · rate differential → carry · calendar → news-flade rules · **ambiguous = all off**
Risk	E5: spread check vs cost floor, session-spread widening, news ladder, kill-switch

Phase	What happens
Venue	cTrader Open API — **50 non-hist req/s**, heartbeat ≤10s, proxy routing (AWS Global Accelerator), NY4

5.2 FX Team 2 (gold + WTI — pure scalping, push mode)

Phase	What happens
Offline (Mode R)	learns this symbol's microstructure: tick patterns, spread rhythm, liquidity windows, session liquidity gaps
Gate	E4 — plus **E7 net-per-attempt cost proof** required (spread + commission per scalp)
Online (Mode T)	executes scalps every tick; maker-entry mandatory; hard tick stop; spread-widening gate; EIA/CPI pre-flatten (oil)
Push rule	R searches hard → E7 proves net edge → **E5 sets size** while net edge positive
Account	**separate account** — scalping risk isolated from Team 1

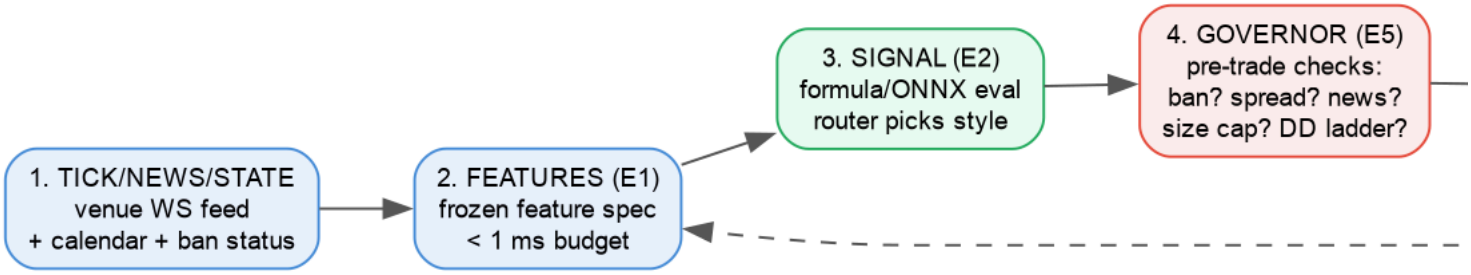
Costs (web-verified, IC Markets Raw): EUR/USD avg 0.1 pip + $7/lot RT; XAU avg 0.07 + $7/lot RT; NY4 servers; scalping allowed. Gold + WTI trade in **both teams, separate money**.

5.3 Crypto (BTC, ETH, HYPE)

Phase	What happens
Offline (Mode R)	funding regimes that mean-revert · cascade precursors (OI + leverage build-up) · weekend dislocation · pair cointegration
Online (Mode T)	features = funding rate, OI, liquidation feed, book depth — **in frozen formulas, never "thinking"**
Exits	cascade-flatten · weekend-gap guard · funding window (00/08/16 UTC) = the print · kill-switch
Venue	Hyperliquid: REST weight **1200/min**, ~100 actions/10s practical, public-API order path measured ~600–700ms end-to-end (unverified: in-house node would cut this)

---

§6 · End-to-end trade lifecycle




```
1 TICK/NEWS/STATE  →  2 E1 FEATURES (<1ms)  →  3 E2 SIGNAL (frozen eval, router picks style)
→  4 E5 GOVERNOR pre-trade (ban? spread? news? size cap? DD ladder?)
→  5 E6 EXECUTE (venue adapter, order + ack)  →  6 E7 POST-TRADE (fill vs expected, slippage + cost log)
→  7 FEEDBACK to research record (nightly)  ⇄ cost model refit → back to 2
```

Exit ladder 1–6 (all markets): 1 pre-flatten before prints ±5min · 2 hard stop (−30–50% premium / tick stop) · 3 time-stop · 4 news-flatten (mid-trade, owner's rule) · 5 liquidity gate on exit (no bid → underlying hedge, close-only) · 6 kill-switch (E5 ladder).

---

§6A · Operations — runbook, kill matrix, recovery (council fixes O1–O6)

The council's verdict: *"the document designs the machine but not the man."* This section closes that gap before delivery.

A.1 Kill matrix (trigger → scope → action → re-arm)

Trigger	Scope	Position action	Re-arm
E5 DD ladder breach	all cells in scope (symbol → team → market → global L0–L4)	**flatten** + cancel resting orders	owner confirmation, cooldown 1 session
Venue feed stale / WS drop	venue's cells	**forced-flat + pause** (no blind trading)	feed healthy + reconcile
Half-sent order / no ack	that cell	hold, reconcile, idempotent client order ID prevents double fill	ledger matches venue
Owner kill (mobile, out-of-band)	global	flatten all, both FX accounts	manual re-arm
Kill confirmation	—	**no confirmation delay** — kill is instant, re-arm is gated	—

Out-of-band backstop (council fix A13): venue-side max-loss order + independent watchdog process that can flatten **without the app** — the L0–L4 in-process cascade must not be the only kill path.

A.2 Startup invariant

`reconcile ledger vs venue positions → then, and only then, accept orders`. Idempotent client order IDs on all 3 adapters (council fix O3).

A.3 Push-mode human guardrail (council fix O5)

Rule	Why
No intraday size-up path — size only via E5, never by hand	governor dies the first manual override
Manual override = cooldown + logged as a counter (feeds E4 review)	overrides become visible data
Manual trades flagged and excluded from E7 labels (or separate sub-account)	owner's own orders would pollute the net-per-attempt proof
Attention budget: threshold alerts + delayed/aggregated P&L view — no live tape staring	Barber & Odean / disposition-effect evidence: tape-watching drives myopic loss aversion

A.4 Daily runbook (minimum)

When	Check
Pre-open	WS health ×3 venues · ban status · news calendar · position-vs-ledger break
In-session	exception-based supervision only — page alerts (tiered: page / same-day / weekly digest; thresholds tuned against replay before go-live, council fix O9)
Post-close	E7 cost reconciliation · E4 promotion queue (**owner-approved, ≤5 decisions/week**, council fix O8) · research batch status
Weekly	drift review · alert-threshold review · venue API/circular drift check · **backup restore-test (E7/G4 append-only + R3 registry)**
Monthly	**kill-drill: feed-stale + venue-down + kill-switch rehearsal** (Session 33 P3)
Always	process supervisor auto-restart · **off-VPS heartbeat** · one-action mobile kill (council fix O10)

---

§7 · Rules and regulations we keep

7.1 Non-negotiables R1–R8 (internal)

#	Rule
R1	no Windows-only runtime in the hot path
R2	deterministic governor — no ML in risk decisions
R3	**LLMs never in hot/risk paths** (R1/R2 rails) — R3 research only
R4	risk first — capped worst case beats extra opportunity
R5	every claim measured vs target vs web-verified vs unverified
R6	no hand-editing generated artifacts / no editing vendored code
R7	one style active per symbol (G5), default = all off
R8	gates external and uneditable (G1–G6)

7.2 Venue regulations (web-verified)

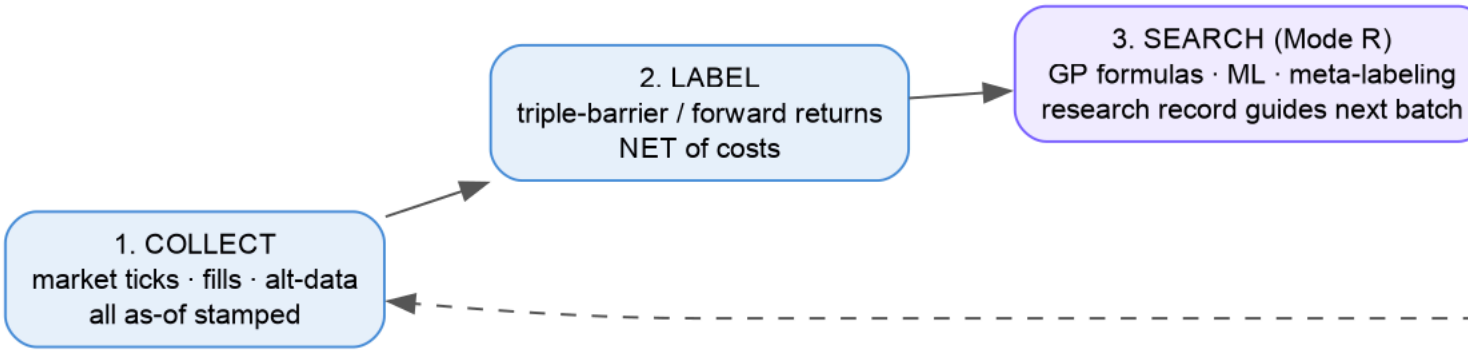
Rule	Source	Where it binds
Algo order tagging — first 12 digits `444444444444` + 13th digit **"0"/"2"/"4"**; all API orders = algo orders; tagging applied broker-side at OMS — confirm Upstox exposes/sets it in W0	NSE/INVG/69255 + corrigendum 69289, NSE FAQ Nov-2025	E6 adapter tags every order; W0 integration test
Rate limits — regular algo **10 rps**, SEBI-registered **50 rps**, 500/min, 2000/30min (**Upstox docs**; NSE May-2025 INVG/67858 = implementation standards, separate doc — council fix V10)	Upstox developer docs	E6 central scheduler caps throughput
Retail algo framework IN FORCE since 1-Apr-2026 (Sep-2025 extension) — not a future deadline (council fixes V1/O11)	SEBI Feb-2025 + extension	**W0 blocker — go-live without it = broker rejects orders**
API access gates: static-IP whitelisting · OAuth-only + 2FA · broker-issued API key · **no market/IOC orders for algo orders** · **≥5-year audit trail** with actual-user ID (council fix V2)	NSE/INVG/67858 §I	E6 adapter + W0 integration test
TOPS 10 OPS = algo-registration trigger (above → register for unique algo ID; below → generic) — **distinct from** Upstox's 10/50 rps API throttle (council fix V5)	NSE implementation standards	E6: throttle ≤10 OPS unless registered
<45 ms order execution — *vendor performance claim (status.upstox.com), NOT a regulation* (council fix V6)	Upstox vendor docs	design target only, §10
MWPL ban >95% → reduce-only — **stock derivatives only** (index options not MWPL-banned); long-standing **exchange** framework (council fix V9)	exchange F&O ban circular	E5 hard entry gate (stock options)
Intraday monitoring ≥4 snapshots/day — **exchange-side duty**; E5 mirrors as **voluntary control** (council fix V8)	SEBI/HO/MRD/TPD-1/P/CIR/2025/41	E5 intraday checks
Option premium paid fully upfront (buy-only = cheap side)	SEBI Nov-2024	sizing math
NIFTY lot = 65 (₹15–20L contract value, Jan-2026)	NSE FAOP70616	position sizing
One weekly index expiry/exchange (NIFTY NSE, SENSEX BSE)	SEBI reform	expiry calendar
FEMA gate (CRITICAL — council fix V3): RBI bars residents remitting margin "for trading in foreign exchange abroad" — **applies to BOTH FX teams**; FX only on SEBI-recognized exchanges / RBI-authorized ETPs. **No India-sourced margin to cTrader until written FEMA/advisor opinion on residence + funding route**	RBI FAQ Q6, Apr-2024 circular	**hard pre-W3 gate**
Crypto gate (council fix V4): Hyperliquid not FIU-IND-registered (status may change); 30% VDA tax + 1% TDS; perp P&L tax position unsettled → FEMA funding review + FIU-IND status check + tax position before any deposit	FIU-IND, 115BBH/194S	**hard pre-W5 gate**
cTrader: 50 non-hist req/s, heartbeat ≤10s; **500 new-orders/min = demo ceiling, production cap to verify**	cTrader Open API docs	Team 1/2 scheduler
Hyperliquid: 1200 REST weight/min, 1 req per 1 USDC traded, 1000 open orders default	Hyperliquid docs	crypto scheduler

7.3 Regulatory non-negotiables R9–R14 (added by council fix V14)

#	Rule
R9	no order without valid exchange algo ID/tag
R10	throttle ≤ TOPS (10 OPS) unless registered algo
R11	no market/IOC orders from algo paths
R12	static-IP-only egress for Upstox
R13	audit trail retained ≥ 5 years
R14	**no India-sourced funding to venues not permitted under FEMA**

---

§8 · Training, self-improvement, self-training (how far it can research itself)



The loop (6 steps, all offline except the live-outcomes feed):

#	Step	Detail
1	COLLECT	market ticks · fills · alt-data — all as-of stamped (R1)
2	LABEL	triple-barrier / forward returns **net of costs** (E7 cost model in the label)
3	SEARCH (Mode R)	GP formula search · ML discovery · **meta-labeling** — the research record guides the next batch
4	GATE (E4)	CPCV (purge = max label horizon + embargo, folds grouped by calendar day across symbols, untouched final holdout) · **DSR ≥ 0.95 vs effective decorrelated trial count** (G4) · crisis replay (EOD index-option history)
5	SERVE	frozen artifact → Mode T (registry R3)
6	MONITOR	rolling IC · PSI drift · live P&L → **retrain: scheduled + drift-triggered**

Self-improvement evidence (web-verified):

- **Meta-labeling (López de Prado):** secondary model on primary signals lifts **PF in 38/42 instruments, DSR in 39/42** (crypto + US equities + FX, daily resolution — **single-source, unreplicated; extrapolation to NSE intraday options unproven**, council fix C14) — the meta-label decides *whether to act* and *bet size*; labels come from the **signal log** (all primary signals, taken or not), not the fill ledger.
- **Automated discovery:** AQuA recursive self-improvement (research-record loop) — Sharpe ~+2.0 walk-forward; QuantEvolve evolutionary multi-agent strategy creation outperforms baselines. **Both single-team arXiv results, no independent replication (council fix C14).**
- **Retraining cadence:** scheduled (daily drift checks / weekly review) + drift-triggered — trigger on **edge degradation (rolling net-of-cost IC vs frozen baseline)**, not raw feature PSI (IV is non-stationary by design → retraining storms, council fix C9); **every retrain+re-gate cycle logged as a trial in G4**.
- **Honesty backbone:** every Mode R trial is logged (G4) → DSR denominator with a **pre-registered search budget capped at 10^3 candidates per style** (council fix C3: raw 10^6 search would make the deflation bar unreachable); failure condition defined up front — *if research can't clear DSR ≥ 0.95 after the capped budget, the scope does not trade.*

What it may NOT do (hard ceiling): self-edit Mode T artifacts (G1), place orders from research (G3), put an LLM in R1/R2 (R3), or enable a style without E4 (R7). "Self-upgrading" = new frozen artifacts through the gate — **never live code rewrite.**

8.5 Technique-coverage matrix (owner Q, Session 29 — "can the agents do all of this by themselves?")

Short answer: yes for ~90% of the quant-technique list — already built in. A few are excluded by the owner's own rules. Caveat: agents run these as *code*, not "understanding."

Technique	Where it lives	How the agent uses it
Monte Carlo simulation	E5 governor + Mode R	worst-case paths before every size; research stress-tests; **fan = swarm harness §8.7**
Walk-forward optimization	E4 gate (CPCV + walk-forward)	automatic in every promotion
Sharpe · Sortino · Kelly · VaR · efficient frontier · CAPM-beta · CLT · Bayes	E4 metrics + E5 sizing	every candidate and live position
Bootstrapping · hypothesis testing · OOS validation	Mode R method 1 + E4	every claim gets a resampled null distribution
Volatility clustering · GARCH	Mode R + E1	candidate models for vol-alpha / gamma cells

Technique	Where it lives	How the agent uses it
Regime detection	**regime router** (rules-only, default off)	computed each tick
Stat-arb · cointegration · OLS · PCA/SVD · Cholesky · Kalman	Mode R method 2 + E1	exactly the stat-arb PAIR stack
Multiple testing	**G4 trial logging + DSR** (honesty backbone)	every trial counted; gate punishes fishing
Multi-timeframe analysis	E1 features	tick → daily bars, as-of stamped
Strategy routing	regime router (Session 26)	enables 0–1 styles/symbol
Adaptive risk	E5 governor	size scales with DD ladder + cost floor
Execution monitoring · trade analytics	E7 ledger	fill vs expected, slippage, per-style cost curves
Portfolio intelligence	L2–L4 + E5	correlation, net exposure, cross-team limits
Black-Scholes · Greeks · binomial tree · finite diff · delta-gamma Taylor · convexity	E1/E5 option math (+ Upstox live Greeks)	verify broker Greeks, recompute deltas for sizing
0DTE · gamma exposure · gamma levels · open interest	option features (OI in §9 data)	features for vol-alpha + gamma-scalper
Intraday seasonality · factor neutrality · regime-dependent momentum	Mode R candidates + E1	researched per symbol, promoted or rejected
Time & sales · footprint · delta · VWAP · orderflow · volume/market profile · absorption · imbalances · auction market theory	**E1 microstructure features** (Team 2 scalping's food)	needs **tick + depth feeds** — §9 gap, procure W0/W4
Heatmap	monitoring dashboards	trivial

Excluded by the owner's own rules:

Technique	Why
Market making	crypto MM deferred (Hyperliquid rebate tier) · equity MM conditional (S1) · options-MM impossible on our venues
0DTE **selling** / short-gamma harvesting	buy-only rule (D4) — 0DTE exists only as **long-gamma buying**
Any of it as ML in the hot/risk path	R2/R3 non-negotiable — ML runs offline in Mode R only

The honest caveat:

"By themselves" means	
Offline (Mode R)	✅ fully automatic — unattended, 10^3 candidates/style cap, logs every trial, may fail
Online (Mode T)	❌ frozen code only — no learning, no improvisation
Promotion	owner-approved, ≤5 decisions/week (council fix O8)
Capability ≠ profit	the agent can *run* any of it — surviving DSR ≥ 0.95 is the gate's job, not the library's
Data dependency	footprint/absorption/time&sales need tick+depth data (§9 gap); historical chains needed for Greeks research

8.6 Memory architecture — episodic · semantic · procedural · working (owner Q, Session 30)

All four memory types already exist in the design — under different names. The honest twist: for non-LLM code agents memory is EASIER and more reliable than for LLM agents, because every "memory" is a file/database, not a model guessing what it remembers.

Memory type	Human/LLM meaning	Fleet equivalent	In design?
Working	scratchpad, current thoughts	live process state: current tick, computed features, open positions, regime state, DD state — inside each Mode T cell + L0–L4	✅ E1/E2/E5
Episodic	"what happened to me before"	**E7 ledger** (signal → fill → slippage → P&L, every trade) + **G4 trial log** (every research attempt, pass or fail) + shadow history + decay events	✅ E7/G4
Semantic	facts & knowledge about the world	**R3 registry**: frozen artifacts · cost model · session/expiry calendars · venue rate limits · ban status · **regime→style map**	✅ R3
Procedural	skills, "how to do things"	**the code itself**: frozen formulas, TWAP/VWAP procedures, exit ladder 1–6, governor rules, §6A runbook	✅ code + §6A

Why this memory beats LLM memory:

LLM memory problem	Fleet answer
Hallucinated memory ("I remember X" that never happened)	**impossible** — episodic = append-only ledger; not in the log = didn't happen
Silent overwrite/decay	frozen + versioned + feature-hashed artifacts (council fix A4); rollback tested
Can't audit a decision	full replay: same as-of data + same artifact = **same decision, bit-for-bit**
Forgets failures, retries them	**G4 counts every failed trial** — the research agent cannot forget what was already tried

Memory write-authority policy (the gap this session fills — enforced, not prose)

Memory	Who may WRITE	Who may READ	Retention
Working	only that cell (its own state)	governor L0–L4	process lifetime
Episodic (ledger, trial log)	**append-only** — E7/G4 only, **nobody edits, ever**	everyone (research feeds on it)	**≥5 years** (R13 audit trail)
Semantic (registry)	**only via E4 promotion** — never the cell itself (G1)	everyone	versioned forever (rollback needs history)
Procedural (code/runbook)	**humans via git + versioned releases — never live-edit by any agent**	everyone	git history

Enforcement (W0b): episodic stores are append-only filesystems/DB roles (no UPDATE/DELETE grants); semantic writes require the E4 signature; procedural = git only, Mode T mounts read-only (council fix A8).

Honest exclusion: vector-store / RAG "agent memory" (the LLM kind) is **not needed in Mode T at all**. It may appear *optionally* in Mode R (R3 rail allows LLM there) — e.g., the research agent recalling past hypothesis notes — but it can only **suggest** candidates; every promotion still passes the deterministic E4 gate.

8.7 Swarm-simulation harness — stolen pattern, deterministic engine (owner decision, Session 32)

The steal (owner: "steal that feature and adopt that too"): MiroFish's orchestration *shape* — seed one scenario → fork thousands of parallel worlds with per-world state → aggregate every world into one report. **The engine is NOT stolen and NOT LLM:** each world is one deterministic math cell (frozen rule + fixed parameter vector + seeded RNG). MiroFish itself was rejected as a dependency (AGPL + LLM swarm = slow, paid, non-replayable); only the pattern is adopted.

MiroFish piece	Our deterministic counterpart
seed info	one hypothesis + scenario manifest (data slice, date range, cost assumptions)
thousands of LLM agents	**N parallel math cells** — same rule, different parameter vector / regime slice (grid or Sobol-sampled)
per-agent memory	each world's carried state (position, cash, style flags)
digital worlds	one seeded-RNG market path or one walk-forward window each
prediction report	**aggregate distribution**: fan of equity curves, percentile bands, worst-case path, survival fraction

Why this is fast: vectorized NumPy/Numba pass — 10^4 worlds × 10 yrs daily ≈ seconds on one CPU, free, **bit-exact replay per world** (seed + parameter hash = world fingerprint, ties to the §8.6 replay-gate idea). An LLM swarm would be ~10^4 inference calls per tick — hours, metered cost, and no two runs identical.

What it computes (Mode R only):

Use	Question it answers
Parameter-ensemble robustness	does the edge survive across the whole neighborhood, or only the one tuned point? (anti-overfit view of E4)
Monte Carlo scenario paths	E5 worst-case inputs — the distribution, not the single backtest
Regime/capacity stress	does it survive every regime slice and doubled size?
O1 predictions exploration	fork plausible futures → count outcomes — **suggest-only**, never a Mode T input

Hard rules (unchanged): worlds live in Mode R (R3) · aggregate output feeds E4 as *evidence*, never as a promotion signal by itself · every world run logged in G4 (counts against the 10^3/style search budget when it explores parameters) · **no LLM inside any world** (R2/R3) · determinism is the point — same seed + same artifact = same fan, always.

---

§9 · Data requirements (what these agents need to train)

Data class	Sources	Used by
1-min+ candles, 6 months history max (Upstox limit)	Upstox candles API	all stock cells (longer history via EOD bhavcopy archive)
Ticks / order book	venue WS (Upstox WS v3: 50 instr/conn; cTrader WS; Hyperliquid book)	scalping, MM, microstructure
Option chain + Greeks (50 instruments/req)	Upstox option Greeks API (iv, delta, gamma, theta, vega, oi)	all 4 option styles
F&O open interest, ban status, expiry calendar	NSE circulars + Upstox	E5 entry gates
Macroeconomic calendar, rate decisions	FRED, central-bank calendars	carry, news-fade, vol-alpha
COT / positioning, news wire	alt-data feeds	Team 1 research, shock-news (S4)
Funding rate, OI, liquidation feed	Hyperliquid info APIs	crypto cells
Own fills / slippage (E7)	internal ledger	cost model, meta-labeling labels

9.1 Data gaps the council flagged (must close before research)

Gap	Impact	Fix before
No historical bid/ask for NFO/BFO (Upstox candles = OHLC only)	E7 would put an *assumed* spread in the label → "net of spread" is circular	W1 — procure historical option tick/chain vendor, or reconstruct EOD spreads from bhavcopy and state the limitation
No historical intraday option chains/Greeks (Upstox Greeks = live only)	vol-alpha + gamma-scalper **unresearchable at operating resolution**	W1 — same vendor; else demote those 2 styles to EOD-resolution research
1-min candles only 6 months back (Upstox)	crisis replay can't run on intraday option history	W0 — build EOD bhavcopy archive; scope crisis replay to EOD index-option history
Stock options monthly + thin → low trade counts	underpowered per-symbol stats (council fix C12)	pool symbols hierarchically; day-1 statistical claims restricted to NIFTY/SENSEX weeklies

---

§10 · Latency and operations-per-second budget

Path	Budget	Evidence class
Tick → E1 features	**< 1 ms P99** (in-process; **IPC fan-out cost measured in W0** — shared feature server feeds all Mode T cells, council fix A1)	design target — unmeasured
E2 signal eval (frozen formula/ONNX)	**< 1 ms P99**	design target
E5 governor checks	**< 0.5 ms** (deterministic, in-memory)	design target
Hot-path total (cell → order)	**< 5 ms P99 internal** + venue network RTT	design target
Upstox order exec	**< 45 ms** — *vendor performance claim, not a regulation* (council fix V6)	web-verified vendor docs

Path	Budget	Evidence class
Hyperliquid public API order round-trip	~600–700 ms → crypto cells scoped **"sub-second", not "every tick"**; in-house node = optional W5 milestone	web-verified
cTrader order path	proxy-routed AWS Global Accelerator; **500 orders/min = demo ceiling — production cap to verify** (council fix V11)	web-verified (demo)
Ops/second — Upstox orders	10/s regular · 50/s SEBI-registered — **central E6 scheduler, token bucket, per-cell fair-share, reserved preemption lane for exits/kill-switch** (council fix A3)	web-verified (Upstox docs)
Ops/second — Upstox market data	50 req/s standard; WS 50 instr/conn → 59 symbols ≥ 2 connections + failover; staleness detector → **forced-flat on stale feed**	web-verified
Ops/second — Hyperliquid	1200 REST weight/min; ~100 actions/10s practical; book WS 100–200ms	web-verified
Ops/second — cTrader	50 non-hist req/s	web-verified
Compute	**75 Mode T + up to 75 Mode R processes** — Mode R gets CPU affinity/cgroup isolation so GP searches can't steal hot-path cycles; throughput table (tick rate × cells + research cores) built in W0 (council fix A2)	design — unmeasured

Quality/optimization levers: one process per Mode T cell (G6, crash isolation) · frozen artifacts = no runtime compilation · **artifact = weights + feature-hash + schema-version; E1 refuses mismatched artifacts; atomic write + tested rollback in W0** (council fix A4) · as-of join once, reuse · co-location: NY4 (IC Markets), Upstox colo, Hyperliquid node later · shadow-first promotion · **startup invariant: reconcile ledger vs venue positions before any order; idempotent client order IDs on all 3 adapters** (council fix O3).

---

§11 · Build plan — waves and time estimates

Wave	What	Est.	Gate to exit
W0a — data foundation	R1 data rail · point-in-time replay engine (leakage tests) · EOD bhavcopy archive · **historical option spread/chain vendor decision**	**4–5 weeks**	data replay passes (no leakage)
W0b — engines + gates	E1/E7 · registry R3 (**versioning + feature-hash + rollback**) · E5 skeleton · **E4 harness (CPCV/DSR/crisis-replay infra)** · E2 router · L0–L4 kill cascade · G5 arbiter · central E6 scheduler (token bucket + preemption lane) · SEBI order-tagging adapter + static-IP/OAuth integration test · **IPC/CPU throughput measurement vs <1ms budget** · signed artifacts + read-only Mode T mounts · **memory enforcement: append-only DB roles for episodic stores (no UPDATE/DELETE grants) + E4-signed semantic writes** (§8.6)	**4–5 weeks**	governor unit tests, rollback tested, IPC overhead measured
W1 — index options (pilot)	NIFTY + SENSEX: 4 buy-styles researched (Mode R) → E4 → 2–4 cells shadow · **swarm-simulation harness (§8.7: seed → 10^4 deterministic worlds → aggregate, vectorized)**	**10–14 weeks** (council fix Q13)	DSR ≥ 0.95 on ≥1 style + 3-mo shadow start
W2 — stock options wave	optionability/liquidity screen (S3) → top-liquid names → hierarchy comms (S4)	**8–10 weeks** (overlaps W1 shadow)	shadow PF > 1 net of costs, hierarchy tests pass
W3 — FX Team 1	11 symbols × styles, cTrader adapter, NY4 VPS	**6–8 weeks** (overlaps W2)	cost floor per style proven in E7 + shadow start
W4 — FX Team 2 scalping	gold + WTI scalping cells, separate account	**4–6 weeks research + 2–3 mo attempt accumulation** (500 attempts don't fit research window, council fix Q13)	net-per-attempt positive over 500+ attempts
W5 — crypto	BTC/ETH/HYPE 4 styles + PAIR (+ optional in-house node milestone)	**6–8 weeks**	funding/cascade cells clear E4
W6 — self-improvement loop	meta-labeling, drift triggers, research record → auto-batches — **moved AFTER W2** (needs ≥6 months of logged signals, council fix Q8)	**4–6 weeks**	drift detector catches injected decay

Mode R toolkit (Session 31 survey, W1 inputs — detail in `Research/notes/SESSION-31-TOOL-EVALUATION.md`): adopt **PySR** (formula search → E2) · **MAPIE** (conformal intervals → E4/E5) · **arch** MCS/SPA (G4 multiple-testing toolkit) · **hftbacktest** fill-model checklist (feed latency + order latency + queue position) as the E4 simulation-honesty reference · **vectorbt blocked pending LICENSE read** (Fair-Code commercial restriction).

Honest calendar (council fixes Q7/A6): shadow windows run **in parallel** with later waves. **"Built" ≈ 7–9 months; "fully live" ≈ 13–15 months** (W1 shadow alone is 3–6 months before its cells go live). Add **30–50% schedule slack** for ops load — from month ~2 the owner is builder + researcher + on-call (council fix O7).

Day-1 = wave 1 = **2–4 live cells**, not 500. Every wave is reversible: artifacts demote via decay monitor **as a state machine: flatten-and-confirm → disable → hand off** — never instant-disable with an open position (council fix A9).

---

§12 · Provenance — what is measured vs claimed

Evidence sub-grades added per council fix Q15: **P** = peer-reviewed/regulatory primary · **R** = preprint · **V** = vendor docs/white paper · **X** = single-source, unreplicated · **E** = extrapolated (proven elsewhere, unproven here).

Claim	Class
75 agents / ≤59 stocks / 11+2 FX / 3 crypto rosters	**Owner-locked** (Sessions 27–28)
Specialist cell per style vs one generalist	**Measured (internal)** — Session 22 ScaleZero (attach artifact in W0)
SEBI 93%/91.1% losers	**Web-verified, P** — SEBI/PwC primary regulatory study
Pillai both-sides-negative	**Web-verified, R** — SSRN preprint
CTA diversification pro	**Web-verified, V** — industry white papers (vendor-adjacent)

Claim	Class
Meta-labeling 38/42; AQuA Sharpe ~+2.0	**X — single-source, unreplicated**; daily US/crypto/FX → NSE intraday options extrapolation **unproven (E)** (council fix Q14)
Venue rate limits, lot sizes, algo tagging	**Web-verified, P/V** — SEBI/NSE circulars (P) vs vendor docs (V)
<45ms Upstox	**V** — vendor perf claim, not regulation
<1ms/<5ms hot-path budgets	**Design targets — unmeasured until W0**
Effective independent bets per optionable symbol	**≈1–1.5, not 4** — long-delta/gamma/vega correlate (council fix Q11); breadth/IR claims re-derived on this number
Buy-only options profitability	**UNPROVEN until Mode R + E4 + shadow** — failure condition: DSR ≥ 0.95 after capped 10^3 budget
In-house Hyperliquid node latency	**Unverified**
Program capital, opex ceiling, tax treatment	**OPEN — owner-set at W0 (§15)**

---

§13 · Open items carried forward

(Pre-council open-item series — distinct from the council ops O1–O12 in §14.)

O1 (predictions trunk 0 cells) · O3 (index F&O granularity) · O4 (7 off-index names) · O5 (styles per instrument — D4 implies 4–9 per optionable symbol) · O7 (hedge-executor per-symbol ×N vs one portfolio cell) · **D4** working spec = OPTIONS-BUYING-SCOPE.md · onboarding optionability screen sets final N.

---

§14 · Validation council (owner requirement — 4 domain agents, adversarial review)

Run 2026-10-27. Four independent reviewers argued against the draft: quant research methods · systems/latency · regulatory compliance · ops/human factors. **60 findings total, 18 CRITICAL.** All CRITICAL findings integrated into the sections above (fix IDs in parentheses); summary:

Lens	Top CRITICAL findings	Status
Quant methods (Q1–Q15)	DSR ≥ 1 impossible as defined (probability) → **fixed to ≥ 0.95** (Q1) · trial denominator unreliable → pre-registered search space + effective count (Q2) · 10^6 search makes gate unreachable → **cap 10^3/style** (Q3) · CPCV unspecified → purge/embargo/grouped folds specified (Q4) · crisis replay infeasible on 6-mo data → scoped to EOD archive (Q5) · **no historical spreads/chains = 2 of 4 styles unresearchable** → vendor added to W0a (Q6) · shadow timeline excluded from 7–9 mo → **split built/live** (Q7)	✅ integrated
Systems/latency (A1–A15)	process-per-cell vs <1ms IPC gap → shared feature server + W0 measurement (A1) · no CPU math → cgroup isolation + throughput table (A2) · E6 throttling stub → central token bucket + kill-switch preemption lane (A3) · R3 no versioning → feature-hash + rollback (A4) · E4/router/kill-cascade unscheduled → W0b (A5) · shadow calendar (A6)	✅ integrated
Compliance (V1–V14)	Apr-2026 already in force → W0 blocker (V1) · missing static-IP/OAuth/audit-trail gates → added (V2) · **FEMA bars India-sourced margin to offshore FX — both teams** → pre-W3 gate (V3) · Hyperliquid FIU-IND/tax → pre-W5 gate (V4) · 45ms = marketing not regulation (V6) · **R9–R14 regulatory non-negotiables added** (V14)	✅ integrated
Ops/human factors (O1–O12)	no operations section → **§6A runbook/kill matrix/startup invariant added** (O1) · kill semantics undefined → kill matrix (O2) · restart/reconcile missing → startup invariant (O3) · shadow contradiction (O4) · **push-mode human guardrail: no manual size-up, manual trades excluded from E7** (O5) · attention budget (O6) · E4 queue owner-capped ≤5/week (O8)	✅ integrated

Open council items (not fixable in-doc, tracked for owner): verify cTrader **production** order limits (V11) · confirm Upstox broker-side algo tagging (V7) · owner to define "order-balance requirement" mentioned in an earlier session (V12, could not identify a regulation by that name) · Hyperliquid FIU-IND status is as-of Oct-2026 and may change (V4).

Council verdict: architecture and honesty framing sound; delivery was blocked on gate math (Q1–Q6), run-ability (A1–A6), legal exposure (V3–V4), and operator safety (O1–O5) — **all now addressed above.**

---

§15 · Business layer — capital, costs, tax, stop-rules (full-plan crosscheck, Session 33)

> Owner asked: "is planning COMPLETE or did we miss anything?" Full re-read of §1–§14 verdict: **the machine is planned; the BUSINESS around it was not.** §15 closes that. One stale number also fixed (§1 cell ceiling recomputed 481–543).

15.1 Capital & account funding (owner sets amounts at W0 — rules are locked now)

Account	Funds	Currency	Funding rule
A1 Upstox	stocks options (≤59)	INR	option premium paid upfront (buy-only = cheap side); tranche per wave
A2 cTrader	FX Team 1 (11 symbols)	USD	**no India-sourced margin until FEMA written opinion (R14, pre-W3)**
A3 cTrader (separate)	FX Team 2 scalping (gold+WTI)	USD	same FEMA gate; **isolated — scalping losses cannot eat A2**
A4 Hyperliquid	crypto (BTC/ETH/HYPE)	USDC	**pre-W5 gate**: FIU-IND status + 30% VDA + 1% TDS position first

Rules: no commingling between accounts · **capital expands only into a wave whose gate already passed** (never pre-fund an unproven wave) · per-wave tranche size = owner decision at that wave's start, recorded in the E7 trial log.

15.2 Opex — the program costs money even with zero positions

Line	Class	When
Historical option chain/spread vendor (§9.1 gap)	one-time setup + subscription	W0a
NY4 VPS (FX), monitoring/backup storage, alerting	monthly	W0b
Alt-data (COT, news wire for S4)	monthly	W1–W3
LLM API for Mode R (optional, R3 only)	metered — **optional, may be zero**	W1+
FEMA/CA advisory (one-time opinions)	one-time	pre-W3 / pre-W5

Rule: E7 cost model must include opex — a style is only "profitable" net of the infrastructure that carries it. Opex ceiling = owner-set monthly number; exceeding it pauses research, never trading risk controls.

15.3 Tax & accounting (India — design obligations, execution = owner's CA)

Item	Regime (web-verified §7.2 where applicable)	Fleet obligation
F&O (stocks/index options)	business income (non-speculative), audited books likely	E7 must export **financial-year P&L + trade archive** in CA format
Crypto	30% VDA on gains + 1% TDS (115BBH/194S)	same E7 export; tax position is part of pre-W5 gate
Offshore FX gains	FEMA route + tax on remitted gains — **unsettled until advisor opinion**	R14 gate; E7 keeps native-currency records ≥5 yr (R13)
Audit trail	≥5 years (R13, NSE requirement)	already enforced (§8.6 append-only)

New W0b build item: E7 fiscal-year export (P&L per account + per style + trade list + deposit/withdrawal log) — the ledger already stores it; the exporter is the missing part.

15.4 Program-level stop rules (new — per-style failure existed, program failure didn't)

Level	Failure definition	Action
Style	DSR < 0.95 after capped 10^3 search	scope does not trade (§8) — exists
Wave	gate not passed in its window	wave stalls; later waves hold (§11) — exists
Program by end of W1 shadow (≈month 7–9): zero styles cleared E4 on ANY market		owner reassesses program (pivot scope/venue/budget) — **new**
Program cumulative realized drawdown > owner-set % of total capital (default suggestion 20%)		global flatten + hard stop until owner re-authorizes — **new**

15.5 Multi-currency reporting (silent-pollution guard)

E7 reports **per-venue native currency** (INR stocks · USD FX · USDC crypto) + INR equivalent at a daily reference rate as a **separate conversion line**. Style statistics are computed in native currency only — otherwise INR/USD drift would contaminate cost curves and the decay monitor would fire on FX noise, not edge loss.

15.6 Pre-go-live verify list (owner/ops items — nothing here is blocked design)

#	Item	Gate
P1	NSE **client position limits** in index options (per-client size ceilings)	before first live cell
P2	Upstox broker-side algo tag confirmation (council V7 = O9)	W0
P3	monthly kill-drill (feed-stale · venue-down · kill-switch) — **added to §6A runbook**	from W0b
P4	**secrets management**: venue API keys in OS keychain/vault, never config files; rotation schedule	W0b
P5	cTrader production order limits (council V11 = O8)	pre-W3
P6	FEMA written opinion (R14)	pre-W3
P7	FIU-IND status recheck (V4)	pre-W5
P8	historical option chain/spread vendor pick (Q6)	W0a

15.7 Completeness scorecard — the honest final answer

Layer	Status	Where
Symbols/styles/architecture/gates/ops/regulatory	**complete, council-validated**	§1–§14
Business layer (capital rules, opex, tax, stop-rules, multi-currency)	**complete as of Session 33**	§15
Remaining unknowns	**W0 measurement items, not design gaps**: IPC latency · opex/capital amounts · vendor pick · broker confirmations	§10, §15.6

Verdict: planning is DONE. What remains is decided by *doing* (W0), not by more planning. The first build action = W0a data foundation.